

Reverse Sobel sequences, and a first glimpse of probabilities

Topics in Language and Mind

3 October 2017

1. An argument for the strict conditional approach: reverse Sobel sequences

Von Fintel and others hear a contrast between the different orders.

- (1a) If the USA thr new its weapons into the sea tomorrow, there would be war; but if all the nuclear powers threw their weapons into the sea tomorrow, there would be peace.
- (1b) ? If all the nuclear powers threw their weapons into the sea tomorrow, there would be peace; but if the USA threw its weapons into the sea tomorrow, there would be war.
- (2a) If Sophie had gone to the parade, she would have seen Pedro. But if Sophie had gone to the parade and been stuck behind a tall person, she would not have seen Pedro.
- (2b) ? If Sophie had gone to the parade and been stuck behind a tall person, she would not have seen Pedro. But if Sophie had gone to the parade, she would have seen Pedro.

Von Fintel's idea: explain the contrast by reference to the idea that quantifier domains can easily expand but not easily contract. So the pairs are analogous to the following pair using 'every':

- (3a) Every philosopher is at the Dean's reception in the Silver Center. But every Chinese philosopher is at a conference in Beijing.
 - (3b) ? Every Chinese philosopher is at a conference in Beijing. But every philosopher is at the Dean's reception in the Silver Center.
- How bad are (1b) and (2b)? As bad as (3b)?
 - Is there any similar contrast between (4a) and (4b)?
- (4a) The closest town with a post-office is an hour away. But the closest town with a post-office and a hospital is five hours away.
 - (4b) The closest town with a post-office and a hospital is five hours away. But the closest town with a post-office is an hour away

2. Moss's epistemic diagnosis

Moss's thought: sometimes, a sentence A makes salient certain hypotheses which we can't rule out with absolute certainty. Once such a hypothesis is salient, we are unwilling to flatly assert things which are inconsistent with it, or which are improbable given it, even when we are very confident that those things are true, and would have happily asserted them in a setting where the hypothesis was not salient.

- (5a) That animal was born with stripes.
- (5b) But cleverly disguised mules are not born with stripes.
- (6a) Cleverly disguised mules are not born with stripes.
- (6b) #But that animal was born with stripes.

According to Moss, 'If Sophie had gone to the parade and been stuck behind a tall person, she would not have seen Pedro' makes salient the following hypothesis, which can't be ruled out with certainty: it's not the case that if Sophie had gone to the parade, she would not have been stuck behind a tall person.

- According to Lewis, this hypothesis can be stated more easily as follows: if Sophie had gone to the parade, she might have been stuck behind a tall person.
- According to Stalnaker, the hypothesis entails the following: if Sophie had gone to the parade, she would have been stuck behind a tall person.
- Note that this hypothesis is not *inconsistent* with the claim that if Sophie had gone to the parade she would have seen Pedro. But that claim is certainly *improbable* given the hypothesis, which is all Moss needs.

It's not crucial that the sentence that does the salient-making is itself a counterfactual:

- (7a) Do you remember when Kate got stuck behind a tall person and missed seeing Pedro in her first baseball parade?
- (7b) ? But if Sophie had gone to the New York Mets parade, she would have seen Pedro.

Does Moss's diagnosis apply to the weapons example? Here the hypothesis is that the following is not the case: if the USA had thrown its weapons into the sea, the other nuclear powers would have held onto their weapons. It's unclear why just uttering the first sentence in the reverse Sobel sequence should make *this* salient in the relevant way!!!

3. A puzzle for von Fintel

Why are the following discourses so obviously fine?

- (8) If all the nuclear powers threw their weapons into the sea tomorrow, there would be peace; but if the USA threw its weapons into the sea tomorrow, the other nuclear powers would not throw their weapons into the sea, and would be war.
- (9) If Sophie had gone to the parade and been stuck behind a tall person, she would not have seen Pedro. But if Sophie had gone to the parade, she would not have been stuck behind a tall person, and would have seen Pedro.

Contrast the following, which looks just as terrible as (3b):

- (10) Every Chinese philosopher is at a big conference in Beijing; but every philosopher is non-Chinese and went to the Dean's reception in the Silver Center.

Tentative suggestion: maybe the oddity of the original Reverse Sobel Sequences (1b) and (2b) is due to the fact that if you were cooperative, you'd say something like (8) or (9) instead? (But why?)

4. An argument against the strict analysis

Setup: yesterday at noon, I was debating whether to toss three coins, but eventually decided not to. Coin 1 was double-Headed; coin 2 was double-Tailed; coin 3 was normal. (Coins of this sort do have a tiny chance of landing on their edges.)

Natural things to say about this setup:

- (11a) It was very likely that coin 1 would land Heads if I tossed it.
- (11b) It was very unlikely that coin 2 would land Heads if I tossed it.
- (11c) There was about a 50% chance that coin 3 would land Heads if I tossed it.

(11c) looks problematic for the strict conditional account!

If the operative accessibility relation for (11c) is one that worlds where I decide not to toss the coin bear both to worlds where Coin 3 was tossed and landed Heads and to worlds where Coin 3 was tossed and landed tails, then the chance at noon that the proposition that coin 3 would land Heads if I tossed it is way less than 50% (maybe even 0%)!

But what sort of accessibility relation could there be such that among the no-toss worlds, some (about half) are such that all the toss-worlds accessible from them are Heads-worlds, and others aren't?